A scale-dependent vacuum

Vacuum energy along the renormalisation scale, one acceleration scale from galaxies to clusters, and a fading dark energy

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Abstract

Vacuum energy estimated at the Planck, electroweak, QCD and cosmic scales differs by up to 120 orders of magnitude. We treat the renormalisation scale μ as a coordinate of a warped geometry and ask how the vacuum changes along it. In that geometry the gravitating vacuum scales as $M_p{}^2\mu^2$ rather than μ^4, so it falls by $(M_p/H_0)^2 \approx 10^{120}$ between the Planck scale and the Hubble scale, and its value at $\mu = H_0$ matches the observed dark energy within 0.3 dex. The same vacuum, partly relaxed around matter, gives a quasi-linear field equation for galaxies whose acceleration scale is fixed at $cH_0/2\pi$. It describes 153 SPARC rotation curves as well as a free scale (ΔBIC = -6.8). On 145 galaxies at $z \approx 0.7$–2.5 the scale does not grow with H(z) (excluded at 3.4σ); a nearly constant scale is consistent but unconfirmed. Adding one pressure scale for hot plasma, measured once on X-COP and applied unchanged to independent data, reproduces cluster masses within 0.15 dex and fills the cores of X-ray groups. Late-time data prefer a fading vacuum ($w_0 = -0.85$, 2.8σ from a constant). General relativity and the Standard Model are recovered wherever tested. All 73 results come from an open test suite with pass/fail rules fixed before running.

Keywords: renormalisation scale; vacuum energy; cosmological constant problem; galaxy rotation curves; galaxy clusters; dark energy

1 Introduction

Every physical structure has a position, a characteristic time and a characteristic size. The first two are coordinates in ordinary physics; size is usually treated as a property. Renormalisation already treats scale as something physics runs along. We take that literally and use the logarithm of the renormalisation scale as a coordinate,

$$S = \frac{1}{k} \ln \frac{\mu}{\mu_0},$$

so that each structure has a place (x, y, z, t, S). In practice a structure's S is the number of e-folds of its size above the Planck length.

The working hypothesis is that the vacuum is a physical medium whose curvature depends on S: negative everywhere, steeper at small S and flatter at large S, and locally relaxed by matter. Gradients in that relaxation then act gravitationally. The paper tests this hypothesis in five places: the geometry it requires (Section 2), the vacuum energy at four scales (Section 3), galaxies (Section 4), groups and clusters (Section

5) and the late-time vacuum (Section 6). Section 7 checks known physics. Section 8 discusses how structures from hadrons to clusters sit along S and how this connects to work on selection in physical systems.

Throughout, "structure" means a bound, localised system at a given scale, and "vacuum" means the medium around it at that scale. The distinction is relative: what is a bound structure at one scale can act as part of the background medium at a larger one (Sections 3 and 5).

2 Geometry

If S is a coordinate, it needs a metric. The simplest metric with S as a direction is warped, of the Randall–Sundrum type [1]. Each identity below was verified symbolically.

$$ds^2 = e^{2A(S)}\left(-dt^2 + a(t)^2 d\vec{x}^2\right) + dS^2$$

- **The vacuum is negatively curved.** Flat slices of fixed S require a negative bulk vacuum, $\Lambda_5 < 0$ (anti-de Sitter).
- **The vacuum seen at a given S** is $\Lambda_{\text{eff}}(S) = \Lambda_5/2 + 3A'(S)^2$. It has a negative bulk part and a positive part from how the geometry bends with S, so it changes with scale.
- **Matter relaxes the vacuum.** Energy on a slice raises A'^2 and offsets the negative bulk term.
- **Sign flip.** Across a slice that carries energy, the bending reverses sign, and the slice's own vacuum term changes sign at a critical energy content.
- **Edges of the observed range.** The lower edge is where the quantum and black-hole lines meet, at $R = \sqrt{2}\,\ell_P$; the upper edge is the Hubble scale, where the universe lies on its own black-hole line. The coordinate itself is not bounded by either.
- **S can move in time.** A homogeneous S(t) obeys consistent coupled equations with the expansion. A ghost-free S(t) cannot make the vacuum grow.

The same geometry fixes four more things:

- **Gravity's strength.** The junction condition at a slice that carries energy gives the slice's own Friedmann equation and its Newton constant, $G_4 = \kappa^4\sigma/48\pi$ [2]. How strongly matter gravitates on a slice is set by the bulk curvature and the slice's energy content.
- **Moving along S is renormalisation-group running.** In the warped bulk $S = (1/k)\ln(\mu/\mu_0)$, so $d/dS = k\cdot d/d\ln\mu$, exactly the RG derivative. Couplings placed along S run as the Standard Model says, with no added term.
- **Gravitational waves travel at c.** A tensor perturbation's zero mode on a slice obeys $\Box h = 0$, so its speed is exactly c.
- **The tuning moves into the geometry.** A slice's own vacuum is the difference between the bulk term and the slice's energy content, and our small positive vacuum sits just above the flip. The cosmological-constant cancellation is carried by the geometry, not removed. In a literal fifth dimension it is one part in about 10^{62}; in the holographic reading the slice tension has no measured value.

The geometry is read holographically, as in the correspondence between a warped bulk coordinate and energy scale [3]. S is scale, not an extra direction of space, so space keeps its three dimensions.

3 Vacuum energy at four scales

Vacuum energy estimated at four scales falls from small to large S: steeply at small S and gently at large S. The table lists the standard estimates.

Scale	μ	**Vacuum energy (eV^4)**
Planck	2.4×10^{27} eV	3.5×10^{109}
Electroweak (Higgs potential)	2.5×10^{11} eV	1.2×10^{44}
QCD (gluon condensate)	2.4×10^{8} eV	3.4×10^{33}
Cosmic (observed Λ)	1.4×10^{-33} eV	2.4×10^{-11}

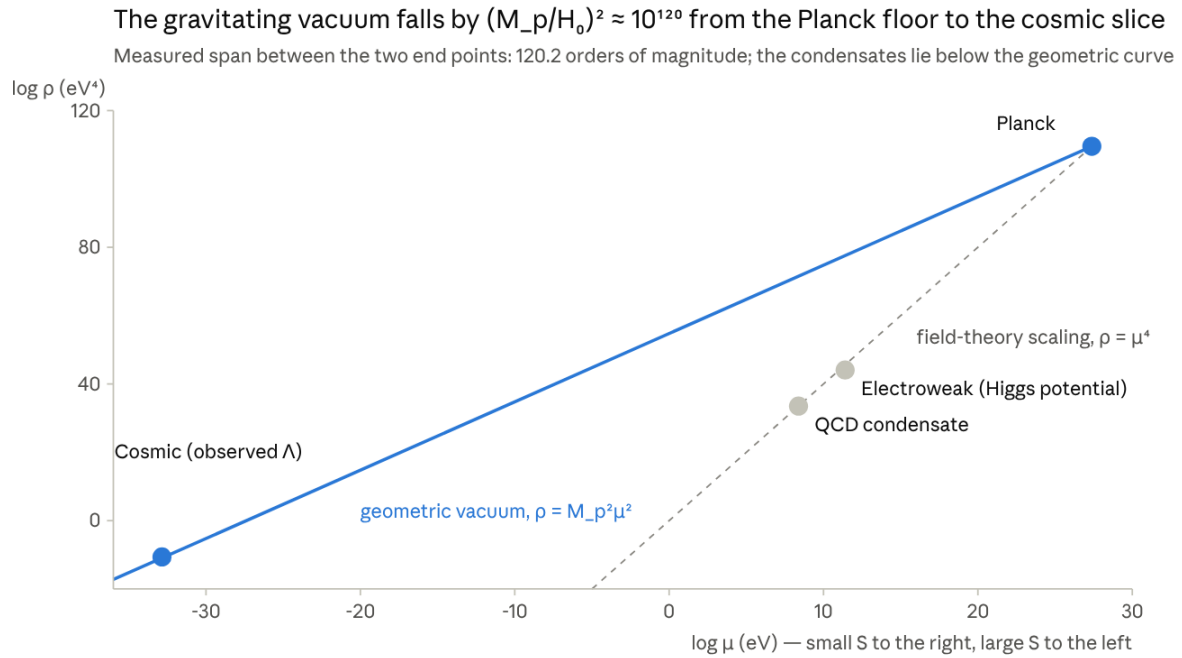


Figure 1 · vacuum values of the table above; curves from Section 2 (test suite, cells p02 and p05)

The gravitating vacuum. On each slice, the vacuum that gravitates is the geometric one of Section 2. Along the exact solution its curvature scales as the square of the slice's energy scale, $\Lambda_{\text{eff}} \propto \mu^2$, so its energy density is

$$\rho_{\text{vac}}(S) \sim M_p^2 \mu^2(S) \qquad (\text{not } \mu^4)$$

At the Planck scale ($\mu = \text{M}_\text{p}$) this is $\text{M}_\text{p}{}^4$, the quantum-field value. At $\mu = \text{H}_0$ it is $\text{M}_\text{p}{}^2\text{H}_0{}^2$, the observed dark energy, within 0.3 dex. Between the two the vacuum falls by

$$\frac{\rho_{\text{vac}}(\text{Planck})}{\rho_{\text{vac}}(\text{cosmic})} = \left(\frac{M_p}{H_0}\right)^2 = 2.9 \times 10^{120}$$

so the size of the discrepancy follows from the geometry. Placing our slice at $\mu = \text{H}_0$ is an identification, not a derivation. The same μ^2 scaling appears in holographic bounds on the vacuum [4]; here it follows

from the geometry, and the same vacuum sets the galactic scale of Section 4.

The condensates. The QCD and electroweak values are not the gravitating vacuum of those slices. They are the vacuum medium of their scales, made of the bound structures around them, and by the sign flip they are carried by those structures: most of a proton's mass is QCD energy. They gravitate as matter, not as vacuum, and they lie far below $M_p{}^2\mu^2$, as they must.

Reading the hierarchy. On this reading the quantum-field estimates and the observed cosmological constant are values of one function at different scales, so comparing them directly mixes quantities from different points on S. Their ratio is the fall of the vacuum along S, $(M_p/H_0)^2$, rather than a failed prediction. The fine tuning does not vanish: it is carried by the geometry (Section 2).

Vacuum and the structures below it. If the vacuum medium at each scale is made of the bound structures just below it, the two should have comparable energy densities:

- at the QCD scale, vacuum energy and nuclear-matter density agree within 0.30 dex;
- at the cosmic scale, dark energy and matter agree within 0.36 dex today.

The cosmic match holds near the present epoch: the ratio of dark energy to matter was 0.29 at z = 1 and 0.002 at z = 9. Why the balance holds now is the coincidence problem, which this reading restates rather than solves.

4 Galaxies: the vacuum gradient

Around a galaxy the vacuum is relaxed where the galaxy's matter dominates and unrelaxed farther out. The gradient between the two acts as unseen mass.

Field equation. Let r(y) be the relaxed fraction of the vacuum where the matter's own pull is $g_N = y \cdot g\dagger$. The unrelaxed part acts through its gradient:

$$\nabla^2\Phi = 4\pi G\rho + \nabla\cdot\left[\left(\frac{1}{r(y)} - 1\right)\nabla\Phi_N\right], \qquad r(y) = 1 - e^{-\sqrt{y}}$$

The unseen mass density is the divergence of the unrelaxed vacuum. The equation has the quasi-linear form of Milgrom [5], and the relaxation profile, the simplest one with the two required limits, matches the observed radial acceleration relation [6]:

- **Where matter dominates,** the vacuum is fully relaxed and gravity is Newtonian.
- **Where the vacuum is unrelaxed,** the response becomes scale-invariant. This follows if a shift along S is a rescaling: in three dimensions the invariant form is $|\nabla S|^3$. It gives $g \to \sqrt{(g_N\, g\dagger)}$, flat rotation curves and $v^4 = GMg\dagger$.
- **Unseen mass grows outward,** from zero inside to $M_{ph} \to \sqrt{(M\ g\dagger)}\cdot r$ outside, following the unrelaxed vacuum.

Data. The equation was tested on 153 SPARC rotation curves [7], marginalising each galaxy's stellar mass-to-light ratio, distance and inclination. With the scale fixed by the cosmic vacuum, $g\dagger = cH_0/2\pi$, the equation describes the sample as well as with the scale left free (ΔBIC = -6.8). The free fit gives 1.06×10^{-10} m/s^2.

The scale. g† is set by the cosmic vacuum, the same vacuum that appears as dark energy (Section 6). That the galactic acceleration scale lies close to cH_0 was noted when it was first measured [8]; here it follows from the scale being set by the cosmic vacuum. Measured values lie at 0.11–0.29 $\times$ cH_0, depending on sample and on the stellar mass-to-light treatment, and rotation curves alone cannot separate $cH_0/2\pi$ from a fixed scale of 1.2×10^{-10} m/s^2: the preference flips with the assumed mass-to-light ratio.

Redshift test. "Set by the cosmic vacuum" has two readings. If the scale follows the expansion rate, it grows into the past:

$$g_\dagger(z) = \frac{cH(z)}{2\pi}$$

reaching 1.8 times today's value at z = 1 and 3 times at z = 2. If it follows the vacuum density, which barely changes over that time, it stays nearly constant. The statistic $R = \ln[g\dagger(z)/g\dagger_{local}] / \ln[H(z)/H_0]$ is 1 for the expansion reading and 0 for the vacuum reading. Dark-matter fractions at R_e come from Price et al. [9]; rotation speeds and baryonic masses at 3 R_e for KROSS [10], KMOS3D [11] and KGES [12] come from the published catalogues [13]. The local scale is measured the same way on mass-matched SPARC discs, and the rule was fixed before running: a reading is favoured when R lies within 2σ of its value and more than 2σ from the other.

Sample	Galaxies	Redshift	R	Favours
Price et al. (dark-matter fractions at R_e)	41	0.7–2.5	$+0.67 \pm 0.23$	expansion
KROSS (rotation and baryons at 3 R_e)	42	0.8–1.0	-0.37 ± 0.65	vacuum
KMOS3D (same)	41	0.7–2.2	-0.87 ± 0.55	vacuum
KGES (same)	21	1.3–1.6	-1.68 ± 1.52	inconclusive
Combined	145	0.7–2.5	$+0.33 \pm 0.20$	expansion excluded; vacuum unconfirmed

Combined, the galactic scale does not grow as $cH(z)/2\pi$: the expansion reading is 3.4σ away and the vacuum reading 1.7σ. The expansion reading is excluded. The vacuum reading, a scale that stays nearly constant over the last ten billion years like the vacuum density, is consistent with the data but unconfirmed: the samples disagree with each other ($\chi^2 = 9.9$ for 3 degrees of freedom), with Price et al. alone pointing to growth. MOND also has a constant scale, but there it is postulated; here it follows from the scale being tied to the cosmic vacuum, whose density barely changes over that time. The data cannot yet tell the two apart; better high-redshift kinematics are needed. High-redshift discs are turbulent and partly pressure-supported, and several catalogue galaxies carry more baryons than their dynamical mass allows, so these systematics are not excluded.

5 Groups and clusters

In a cluster the same vacuum gradient acts at two scales at once: around each galaxy, and around the cluster as a whole. Two effects that are silent in a single disc galaxy then come into play.

The galaxies' own vacuum profiles. A cluster's galaxies are the bound structures one scale below the cluster. Each keeps the unrelaxed-vacuum profile it has in isolation, out to the radius where it stops standing out from the surrounding matter: $\rho_v(r) = \sqrt{(M\, g\dagger/G)}/(4\pi r^2) = \rho_{matter}$. Counted over the measured galaxy population (stellar mass function [14], stars distributed with the measured concentration [15], measured stellar fractions [16, 17]), these profiles supply the outer parts of clusters with no adjustable quantity.

Hot plasma. At 10^7–10^8 K the gas of groups and clusters is fully ionised. Its atomic structure is gone, so at the galactic scale it no longer acts as bound structure and instead acts as part of the background medium: by the sign flip it adds to the unrelaxed vacuum instead of relaxing it. Its thermal pressure therefore keeps the vacuum unrelaxed:

$$y_{\rm eff} = \frac{y}{1 + P_{\rm th}/P_0}$$

where matter is bound (disc galaxies, cold groups) $P_{th} = 0$ and the galactic law of Section 4 is unchanged. The pressure scale is measured once, from the 12 X-COP clusters [17]: $P_0 = 1.1 \times 10^{-12}$ Pa.

Results. With P_0 fixed, the mechanism was applied to systems not used to measure it.

System	Data	Residual (dex)
X-COP clusters, 12 (hot-gas masses)	X-COP [17]	−0.08 / +0.07 / +0.06 / −0.08 (core, 200–500, 500–1000, >1000 kpc)
Clusters, 10 (galaxy-kinematic masses)	Li et al. [18]; pressure from the universal profile [19]	+0.06 / +0.14 / +0.07 / −0.05
Relaxed X-ray groups, 15, cores ($r < 0.2\ r_\Delta$)	Gastaldello et al. [20]	+0.19 without the plasma term → +0.01 with it
Central cluster galaxies, 50 (MaNGA)	Tian et al. [21]	offset above the galaxy relation: observed +0.47 / +0.29 / +0.07, predicted +0.45 / +0.32 / +0.19

Cluster masses are reproduced within 0.15 dex in every region of two independent datasets, built with different methods (X-ray hydrostatic and galaxy kinematics). A stricter 0.1 dex criterion, set before running, is met everywhere except the 200–500 kpc band of the kinematic set (+0.14 dex), where the two datasets disagree with each other by up to 0.2 dex for the same cluster (A2029). Group cores are filled, and central cluster galaxies sit above the galaxy relation by the amount their hot surroundings predict, running high at the strongest pull. The galaxies' own profiles carry the outskirts; the plasma carries the cores. The pressure scale P_0 is measured, not derived, so whether the plasma term is a physical mechanism or a single parameter flexible enough to absorb the cluster residuals remains open; what supports it is that one P_0, fixed on X-COP, carries over unchanged to independent kinematic clusters and X-ray groups.

6 The late-time vacuum

At large S the geometry requires the vacuum to flatten toward zero, and a ghost-free S(t) cannot make it grow. Current data agree: dark energy is fading today.

Data and result	Value
Dark-energy equation of state today, w_0 (SN + BAO + CMB + chronometers)	−0.85; constant vacuum disfavoured at 2.8σ
S(t) moving in the potential the geometry fixes	fits as well as a constant vacuum, $w \approx -0.95$ today
Long-term behaviour of that S(t)	vacuum energy tends to zero

The data are Pantheon+ supernovae [22], DESI DR2 BAO [23], Planck 2018 distance priors [24] and 17 cosmic-chronometer H(z) measurements (Moresco et al., as compiled in [25]), each with its published uncertainties.

The vacuum that sets the galactic scale in Section 4 and the vacuum fading here are the same vacuum at two scales: the unseen-mass effect and dark energy are the attractive and repulsive faces of one

scale-dependent vacuum.

Two limits apply. The fitted history behind $w_0 = -0.85$ has dark energy below $w = -1$ before $z \approx 0.4$, which a ghost-free S(t) cannot follow: the data agree with the fading today, while the model's own S(t) fits them as well as a constant vacuum, not better. And S(t) must start at today's dark-energy density, so it moves the "why now" coincidence into its initial condition rather than removing it.

7 Null tests

Wherever general relativity and the Standard Model are already tested, the model recovers them.

Null test	**Result**
Gravitational-wave speed	exactly c: the graviton zero mode obeys $\Box h = 0$ (Section 2); GW170817 allows $-3 \times 10^{-15} \leq \Delta c/c \leq 7 \times 10^{-16}$ [26]
Atomic clocks	no drift predicted, since atomic couplings sit at their own scale; measured $(1.8 \pm 2.5) \times 10^{-19}$ per year [27], 0.7σ from zero
Running couplings	moving along S is renormalisation-group running (Section 2), so Standard-Model running holds
Short-distance gravity	the holographic reading leaves inverse-square gravity unchanged; a literal fifth dimension would be bounded below 39 μm [28]
Nucleosynthesis	vacuum running acts as at most $\Delta N_{\text{eff}} \approx 0.02$
Growth of structure	BOSS $f\sigma_8$ [29]: $\chi^2 = 1.24$ vs 1.28 for ΛCDM
Densities	every measured density lies at least 78 orders of magnitude below the Planck density
Black holes	a Planck-density core sits inside the horizon of every black hole above 4×10^{-9} kg; exteriors are unchanged from general relativity

8 Discussion: structures along S and selection for persistence

Placing known structures on the S axis gives a check on the coordinate itself and connects the model to recent work on selection in non-living systems. S runs from 0 at the Planck length to 139.9 e-folds at the Hubble scale.

Structure	**S (e-folds)**	**Position on the axis**	**Density log ρ (kg/m^3)**
Proton	45.1	32%	17.8
Uranium nucleus	47.2	34%	17.4
Hydrogen atom	56.1	40%	3.4
Self-replicating RNA	60.6	43%	2.3
Virus	63.0	45%	3.3
E. coli	65.9	47%	2.4
Human	79.7	57%	1.4
Giant sequoia	83.6	60%	0.7
Neutron star	89.2	64%	17.6
Earth	95.4	68%	3.7
Jupiter	97.8	70%	3.1
Sun	100.1	72%	3.1
Globular cluster	120.0	86%	−16.8
Milky Way	128.5	92%	−21.7

Structure	S (e-folds)	Position on the axis	Density log ρ (kg/m^3)
Coma cluster	131.5	94%	−22.8

Stability limits. Two lines bound the map. The quantum line, R = $\hbar$/Mc, sets how small a mass can be confined. The black-hole line, R = 2GM/c^2, sets how compact it can be. They meet at the Planck length. All 22 catalogued structures lie inside the triangle they enclose.

Tiers from the couplings. Running couplings place each tier boundary where it is observed:

Boundary	Predicted	Observed	Offset (dex)
Hadron size, where $\alpha_s = 1$	6.9×10^{-16} m	8.4×10^{-16} m	0.09
Nuclear density line	1.2×10^{18} kg/m^3	3.9×10^{17} kg/m^3	0.49
Condensed-matter density line	2.7×10^{3} kg/m^3	4.3×10^{3} kg/m^3	0.20
Largest cold body, $\sqrt{(\alpha/\alpha_G)}\cdot a_0$	5.9×10^{7} m	7.0×10^{7} m	0.07
Stellar mass, $\alpha_G{}^{-3/2}\, m_p$	3.7×10^{30} kg	2.0×10^{30} kg	0.27

Matter and particles cluster on two density lines, nuclear (log ρ = 17.6) and condensed (log ρ = 2.4). Each line is set by the force that dominates that stretch of S: the strong force below the hadron boundary, electromagnetism above it. Gravity overtakes electromagnetism at the planet–star boundary. The tier boundaries agree with standard scaling arguments [30, 31]; what the coordinate adds is a single axis on which all of them can be read together.

Self-replicating systems. Replication needs chemistry, which requires sizes above about 10 a_0, and it needs freedom from gravitational crushing, which requires sizes below $\sqrt{(\alpha/\alpha_G)}\cdot a_0$. That window spans S = 58–98. Every catalogued self-replicating system, from self-replicating RNA to the largest organisms, lies at S = 61–84, inside it.

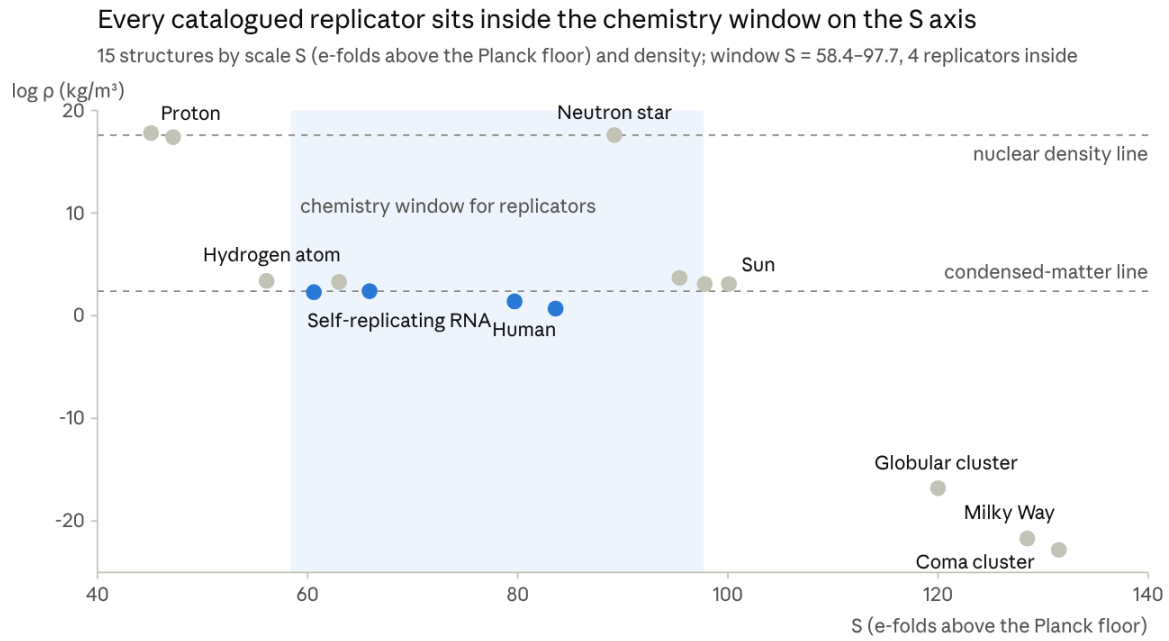


Figure 2 · structures of Section 8 (test suite, cell p03) · window from 10 a_0 to $\sqrt{(\alpha/\alpha_G)}\cdot a_0$

Selection for persistence. Wong et al. [32] propose that stars, minerals and living systems all evolve by selection for persistence: configurations that remain stable are retained, and new ones are explored

from them. The stability limits and tier boundaries above can be read the same way, as the conditions that decide which structures persist at each S. Two features of the model fit this reading. First, the bound structures of one tier act as the background medium of the next: hadrons for nuclei, the QCD condensate inside nucleons (Section 3), and hot plasma at the galactic scale (Section 5). What counts as structure or medium is set by the scale it is viewed from. Second, vacuum fluctuations sit between bound structure and medium, and only some become real particles under the right conditions. Variation at small S and retention under stability limits then give a successive stabilisation of more complex structures as S increases, with self-replicating systems confined to the window above. Smolin's cosmological natural selection [33] applies a related selection argument to the vacuum itself. Developing this reading quantitatively is left to future work; none of the results of Sections 3–7 depends on it.

9 Conclusion

Treating the renormalisation scale as a coordinate of a warped geometry gives a vacuum that changes along it. Measured at four scales, that vacuum falls in the order the geometry requires, and its gravitating part falls by $(M_p/H_0)^2$ from the Planck scale to the Hubble scale. Around galaxies its gradient reproduces the rotation of disc galaxies with a strength set by the cosmic vacuum; in groups and clusters the galaxies' own vacuum profiles and the unrelaxed vacuum of hot plasma reproduce the observed masses; and at late times it is fading. Known physics is recovered wherever it is tested, and on 145 high-redshift galaxies the galactic scale does not grow with the expansion rate; a nearly constant scale remains consistent but unconfirmed.

Data and code

All 73 results come from an open test suite with an executed notebook, run end to end on public data, downloaded with their checksums recorded. Every result is labelled: a derivation, a test with a rule fixed before running, a measurement, a constraint, a diagnostic, or a stated prediction. High-redshift rows are included in the repository, each checked against its catalogue. The X-COP cluster profiles and the cluster data of Li et al. are taken from the compilation in the public repository of R. Monjo [34]. Nothing the code computes is discarded. Two earlier formulations, a linear galaxy law and a single vacuum curve through all four vacuum values, are kept with their pre-set rules and outcomes and marked as superseded by Sections 4 and 3.

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Data and code availability. All data, code, the executed notebook and the results file are openly available

at https://github.com/THEREALMIKEONE/A-scale-dependent-vacuum-Repository.

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